

Sinusoidal Gust Response of RC Propellers in Tandem Configuration

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ABSTRACT

The study of the fixed-pitch small-scale propeller under sinusoidally time-varying freestream is extended to a tandem propeller configuration. The changes in thrust and pitching moment of both the front and back propeller were characterized when encountering the sinusoidal gust in edgewise flight conditions. The propeller's steady freestream performance obtained in the previous study was used to predict response under sinusoidal time-varying freestream in edgewise flight. Below a reduced frequency of 0.2, the propeller response agrees with the prediction model, suggesting a linear superposition and quasi-steady dynamics for the propellers. At a higher reduced frequency, a reduction in mean thrust and pitching moment was measured. A phase lag in the propeller response was also observed indicating an unsteady response of the propeller. When placing the propeller in close proximity to each other, additional variations in the back propeller's performance were also observed at higher reduced frequencies, creating additional instability in the system.

NOTATION

C_T	=	Propeller thrust coefficient; $C_T = \frac{T}{\rho n^2 D^4}$
C_P	=	Power coefficient; $C_P = 2\pi C_Q = \frac{P}{\rho n^3 D^5}$
C_Q	=	Propeller torque coefficient; $C_Q = \frac{Q}{\rho n^2 D^5}$
D	=	Propeller diameter, (m)
J	=	Advance Ratio; $J = \frac{V_\infty}{nD}$
k_ω	=	Normalized Shuttering System Frequency
n	=	Propeller rotational speed per second,
P	=	Power, (W)
Q	=	Torque, (Nm)
T	=	Thrust, (N)
V_∞	=	Freestream Velocity, (m/S)
ρ	=	Air density, (kg/m^3)
θ	=	Propeller incidence angle, (deg)
γ	=	Propeller pitch, (m)
σ	=	Propeller solidity; $\sigma = \frac{s}{0.25\pi D^2}$
ω	=	Shuttering System Frequency (Hz)
Ω	=	Wind Tunnel Fan Rotational Speed (RPM)
$\Delta\phi$	=	Phase Lag (rad)

INTRODUCTION

With the increasing usage of fixed-pitch-propeller-driven unmanned aerial vehicles (UAVs, or drones), more investigation has been conducted on the performance of fixed-pitch propellers in edgewise flight. Gust encounters or aggressive maneuvers add the complication of time-varying

relative freestream, where not only is there a variation in blade sectional velocity magnitude and angle from advancing/retreating blades, but also from the change in the oncoming freestream itself.

A large literature on helicopter rotor unsteady response is motivated by wake interaction between blades [Ref. 1-3], and by flapping, fluttering, or other issues pertaining to rotor dynamics. In the current study, we consider time-varying freestream velocity or gust encounter of a pair of RC propellers in a tandem configuration. Such unsteady conditions cause a delay in response to control input as well as changes in power consumption [Ref. 4-6]. Unsteady effects caused by the multi-rotor tip vortex interaction, wake from the drone structure, and external payload also affect the performance of the multi-rotor drone, especially in higher speed forward flight [Ref. 7].

Time-varying freestream has also been studied on helicopter rotors theoretically and via CFD [Ref. 8-11]. Most of these theories are an extension of Theodorsen's unsteady airfoil theory [Ref. 12]. Since the drone rotors are relatively rigid, concerns of flapping/fluttering as in helicopter blades [Ref. 1] do not directly apply. While both helicopter and propeller blades experience azimuth-varying loading in nonzero relative freestream, fixed-pitch propellers lack cyclic control or passive cyclic degrees of freedom.

In our previous study [Ref. 13], two fixed-pitch APC propellers [Ref. 14], APC 11x7 and APC 14x14, were tested at different incidence angles (θ) between 0 and 90 degrees, under a normalized sinusoidal gust frequency ($k_\omega = \frac{b}{V_\infty} \omega$)

up to 0.196. The steady freestream performance data under various incidence angles were used to build a steady-state model to compare with measurements of coefficients of thrust, power rolling moment, and pitching moment with respect to the propeller axis under the sinusoidal gust. A good fit between model and measurement is found for the lower pitch-to-diameter propeller which experiences minor blade stall, especially at higher incidence angles where the propeller is experiencing the edgewise flight.

The current study expands our previous study to a tandem rotor configuration at a higher reduced frequency, which is a better representation of a popular propeller-driven drone application. Rotor aerodynamics in tandem configuration has also been widely studied in helicopter configurations. When compared to a coaxial rotor configuration, tandem configuration provides better hovering efficiency when placing the propeller $1.1 d/D$ apart [Ref. 15-17]. However, with the increment of forward flight speed, the velocity induced by the front rotor will affect the inflow of the back rotor and reduces its thrust generation and efficiency [Ref. 16 Ref. 18-19].

With a much smaller rotor size for the propeller-driven drones/quadrotors allowing designs with increased rotor distance, many studies have been conducted to understand the optimum propeller distance in tandem configuration [Ref. 20-22]. Ref. 20 and Ref. 21 found that placing the propellers $3.0 d/D$ apart renders the interference negligible. However, such rotor placement might not be feasible for some of the quadrotor designs with large rotors, resulting in an unavoidable interference between the propellers, and between the propeller and quadrotor fuselage.

Our previous study [Ref. 19] investigated the tandem rotor performance in steady freestream. Up to a 25% increment in power required was observed when compared to a standalone propeller with the same thrust generation over an advance ratio of 0.2. A decrement in thrust coefficient and increment in power coefficient was found for the back rotor. Increasing the vertical or horizontal displacement, or decreasing the incidence angle of the propeller would result in a reduction in the power required due to less interference between the front and back propeller.

Ref. 23 studied the quad-rotor performance when encountering a side wind gust. A change in drone thrust up to 50% is measured when encountering a side gust similar to the quadrotor forward flight speed. Ref. 24 studied the unsteady response of a DJI 9450 quadcopter under a pitching motion up to 1Hz motion frequency. An unsteady response of the quadcopter pitching moment was observed between 75° and 105° incidence angle. The interaction between the front and back rotor, especially the tip vortex interference when pitching the quadcopter over 90° resulted in an additional thrust loss of the back rotor, decreasing the overall stability of the quadcopter.

Due to the complexity of the tandem rotor interaction under steady conditions, it is necessary to investigate the changes in propeller performance under a sinusoidal gust, especially the changes in the back rotor. The study will

provide a better understanding of the propeller performance and drone stability under gust encounters.

EXPERIMENTAL SETUP

Force Based Experiment

Experiments were conducted at the University of Dayton Low-Speed Wind Tunnel (UD-LSWT) in the open jet test-section configuration. The wind tunnel has an inlet contraction ratio of 16:1, a 762 mm x 762 mm testing section inlet, and a 1.37 m x 1.37 m collector. The freestream range is from 1.5 m/s to 20 m/s, resulting in an advance ratio, J , between 0 and 0.5. The turbulence intensity of the open jet configuration is 0.1% at 15 m/s measured by a hot wire anemometer.

A sketch of the experimental setup is shown in Figure 1. The KDE CF-125 propeller is paired with KDE2814-XF-515 brushless motor and Hobbywing Platinum V4 electric speed control, then attached to an ATI Industrial Automation Mini-40 six-component force/torque transducer [Ref. 25]. The transducer has a maximum of 40N on its x and y-axis, 120N on its z-axis, and maximum torque of 2 Nm for all axes. Thrust is the axial force along the z-axis of the sensor, and normal force is along the y-axis. The motor torque is the moment of the z-axis, while the propeller pitching moment is the torque on the x-axis, and the propeller rolling moment is the y-axis torque. The F/T transducer is mounted on a 3D-printed transducer interface, which is then bolted to an 80-20 aluminum extrusion. The extrusion is connected to a Fuyu FSL-40 linear traverse by a 3D-printed traverse interface, which can be replaced to change the propeller incidence angle. The linear traverse is placed at the centerline of the test section and is capable of changing the vertical distance of the propeller. The propeller incidence angle is adjustable, which is to say, the propeller pair can be tilted with respect to the tunnel freestream. The entire assembly is then mounted to a 1.22 x 0.92 x 2.44 meter aluminum frame, secured to the lab floor and the inlet of the wind tunnel.

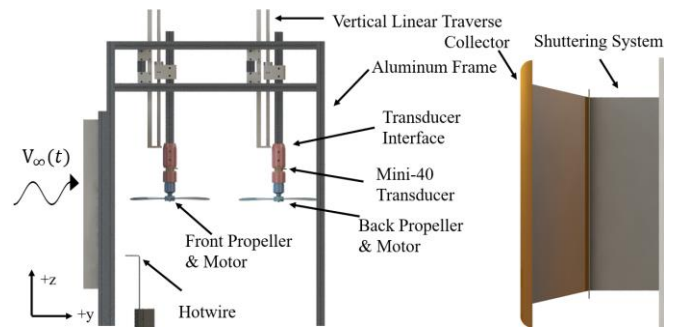


Figure 1. Schematic of the experimental setup

The test matrix is shown in Table 1. The KDE CF-125 propeller is selected based on its popularity in modern drones, and a relatively good performance in edgewise flight. The front propeller rotational speed is set at 66 and 75 RPS, which provides a good signal-to-noise ratio. The peak Reynolds number is calculated at 75% blade span accordingly. To achieve different advance ratios, the front propeller RPS is

fixed, while the wind tunnel freestream velocity is varied. 75, and 90 degree incidence angles were chosen, representing realistic operational range for drones in hovering/transition to forward flight and forward flight. The distance between the front and the back propeller is calculated based on the distance of the propeller hub, as shown in Figure 2. To achieve the trim condition in forward flight, the tandem rotor pitching moment is balanced. In this case, the front propeller rotational speed is fixed, while the back propeller rotational speed is increased so that the normalized system pitching moment, $C_{Qx\text{ norm}}$ (Eqn. 1) is less than 5%. The trim condition is achieved before the tandem propeller system encounters the gust.

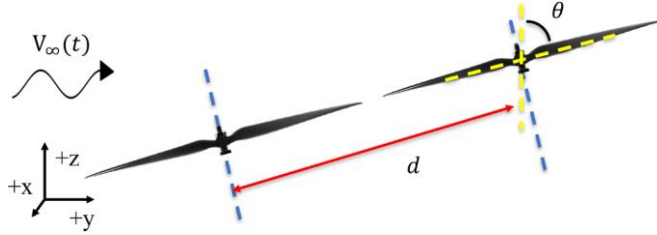
$$C_{Qx\text{ norm}} = \frac{C_{QF} + C_{QB} + (T_{\text{Front}} - T_{\text{Back}}) * \frac{0.5d}{\rho n^2 D^5}}{C_{Q\text{ Front}} + C_{Q\text{ Back}}} \quad (1)$$


Figure 2. Schematic of propeller setup

Table 1. Testing Matrix

Testing Condition	Value
Propeller	KDE CF-125
Diameter	0.318 m
Number of Blades	2
Front Rotor Rotational Speed	66, 75 RPS
Back Rotor Rotational Speed	66 - 115 RPS
Peak Propeller Reynolds Number	160,000
Freestream Velocity	5 - 15 m/s
Advance Ratio	0.1 - 0.5
Propeller Incidence Angle	75, 90 Degrees
Propeller Downstream Distance	1.25 - 1.75 d/D
Reduced Frequency	0.01-0.3

A Dantec Dynamics [Ref. 26] P5516 Probe CTA hotwire is mounted more than 2 propeller diameters upstream of the propeller at the centerline of the wind tunnel testing section inlet, 152 mm above the bottom of the inlet. The hotwire was used to measure the instantaneous freestream u-component of velocity. The hotwire is connected to a Dantec Dynamics 54T30 Mini-CTA and a National Instruments USB-6259 DAQ [27]. The hotwire is calibrated from 0-20 m/s, measured by a TSI 5825 Micro anemometer. A Thor Labs 5 mW laser and a DET10A photodiode are used to measure the propeller rotational speed, also connected to the NI-USB-6259 DAQ. Both hotwire and photodiode are sampled at 12,500 Hz.

The Aerotech shuttering system is installed downstream of the test section, between the collector and diffuser, as seen in Figure 1. The system consists of a set of primary counter-rotating flat-plate vanes mounted in a plane normal to the flow, and a pair of side louver vane sets mounted in the sidewalls of the diffuser. The primary louvers operate over a louver angle range of 41° from full open flow to closed flow (shown

in Figure 3), while the secondary vanes are intended to minimize large-scale flow separation. The shuttering system is operated by a National Instruments PCI-7340 motion control card, controlled by LabVIEW at a system frequency of 125 Hz. The shuttering system is characterized up to 2.5 Hz under a sinusoidal time-varying freestream, corresponding to a maximum normalized gust frequency of 0.3 for the propeller at a tunnel fan speed (Ω) of 300RPM.

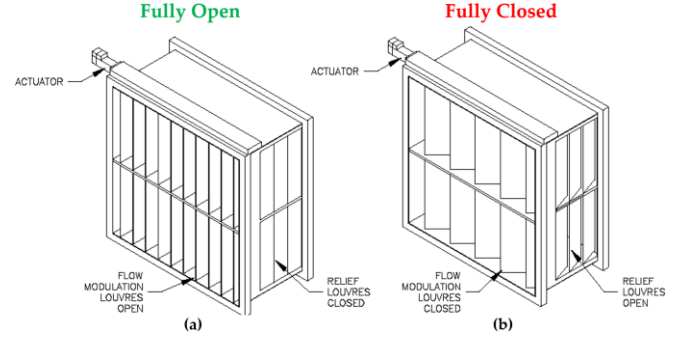


Figure 3. Shuttering system primary louvers (a) fully open; and (b) fully closed

An example of the sinusoidal freestream generated by the shuttering system at 2.25 Hz and $\Omega = 200$ RPM is shown in Figure 4. The details for the characterization are discussed in our previous study [Ref. 14]. The shuttering system linear actuator encoder position, F/T transducer, hotwire, and photodiode signal were all time synchronized for data acquisition.

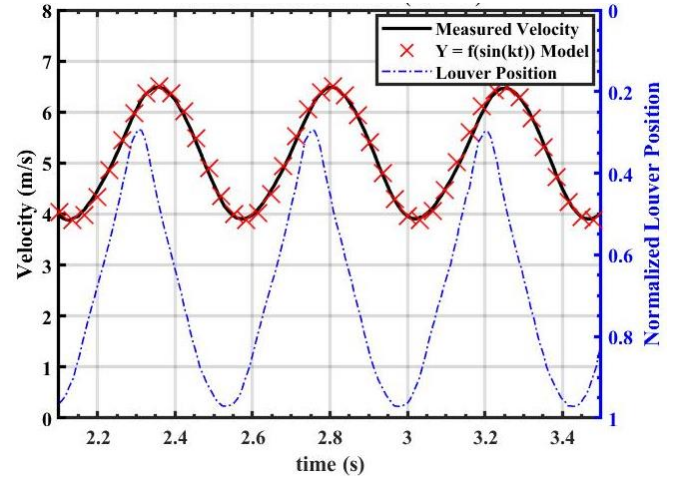


Figure 4. Example freestream velocity at $\omega = 2.25\text{Hz}$ and $\Omega = 200$ RPM

RESULTS

A. Tandem Propeller Performance in Steady Freestream

The tandem propeller performance under a steady freestream was investigated and used to obtain a quasi-steady model. Two different front propeller rotational speeds between 66 and 75 RPS were tested to account for the Reynolds number effects and check for repeatability. As the difference in the experimental result of thrust and power

coefficient between the lowest and highest RPS was less than 3.5%, the result for the front propeller at 66 RPS is selected and discussed in this section.

As mentioned previously, the tandem propeller configuration is tested under approximate pitch trim configuration, which requires the back propeller to operate at a higher rotational speed to generate a higher thrust force to balance the increasing pitch-up moment generated by both propellers based on the experimental result from our previous study [Ref. 19]. The thrust and pitching moment of each propeller is measured at each advance ratio tested, the pitching moment of the tandem rotor system is then calculated based on the summation of the pitching moment from each propeller and the pitching moment generated by the propeller thrust force. The trim condition is achieved when considering a normalized system pitching moment, $C_{Qx\text{ norm}}$ (Eqn 1) less than 5% at each advance ratio tested.

The ratio between the rotational speed of the back propeller (n_b) and the front propeller (n_f) vs. the front propeller advance ratio at a front propeller rotational speed of 67 RPS are shown in Figure 5 for all d_y and θ cases tested. An increment in the back propeller RPS is found with the increment in J for all cases. At $J \sim 0.5$ the rate of back rotor increments in rotational speed plateaus. This is due to the changes in the propeller C_{Qx} performance trend at $J > 0.5$, which is discussed in our previous study [Ref. 19].

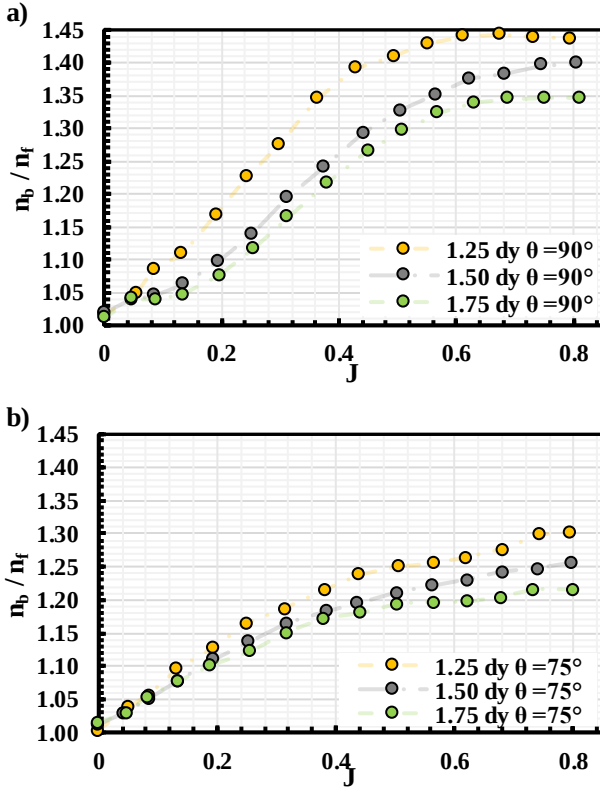


Figure 5. The RPS ratio of back to front propeller vs. J at front propeller of 67 RPS. a) $\theta = 90^\circ$ and b) $\theta = 75^\circ$

With decrease in d_y , the influence of the front propeller on the back propeller increases. Moreover, a smaller moment

arm between the propellers occurs at smaller D_y . The rotational speed of the back propeller has to increase further to generate the required thrust to achieve the trim condition. This trend holds for both θ cases tested. It is also worth noting that at $\theta = 75^\circ$, the back propeller is operating at a lower rotational speed when compared to the one under the same testing condition (J, d_y) at $\theta = 90^\circ$. This is due to the lower C_{Qx} generated by both propellers when reducing θ , which is also discussed in detail in our previous study [Ref. 19]. In this case, a lower thrust generation is required for the back propeller, and hence, a lower n_r is required.

The experimental result for the front propeller C_T and C_{Qx} at $\theta = 90^\circ$, along with the single propeller (marked in black) which serves as a baseline, are shown in Figure 6. The change in C_T is nearly constant for $J < 0.2$, and then rises linearly at $J > 0.2$. For $d_y = 1.25$, a $\sim 5\%$ increment in C_T when compared to the single propeller is observed across the range of J tested, as a result of the positive effect of the back propeller, which is also observed in other studies [Ref. 21] and our previous study [Ref. 19]. This benefit of the front propeller C_T decreases as d_y increases. The trend in C_{Qx} is very similar to C_T where a higher C_{Qx} is observed for a smaller d_y . The trends in both C_T and C_{Qx} for the front propeller holds for $\theta = 75^\circ$ cases.

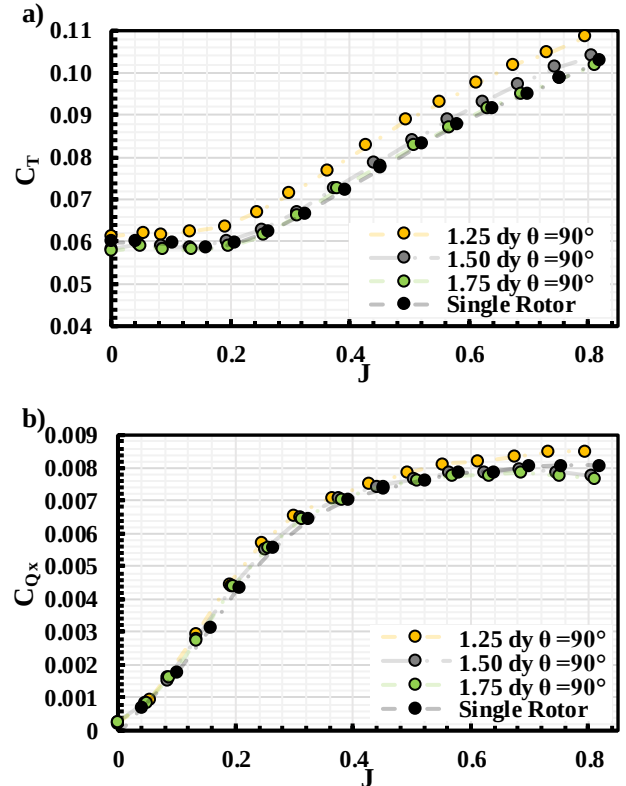


Figure 6. Front Propeller a) C_T and b) C_{Qx} vs. J at various d_y and $\theta = 90^\circ$

Back propeller C_T vs. J at $\theta = 90^\circ$ and 75° are shown in Figure 7. Since the back propeller operates at a higher rotational speed than the front propeller at the same V_∞ , the

resulting J for the backpropeller is lower than that of the front propeller. Similar to the results in Figure 6, there are no discernible variations for $J < 0.2$. At $J > 0.2$, there is generally a decrease in C_T for all cases when compared to the single rotor performance. The decrement in C_T is higher for the smaller d_y case, as a result of higher propeller interactions. It is worth noting that at $\theta = 75^\circ$, the variation in the back propeller C_T when increasing J is relatively small due to the nature of the propeller performance at this incidence angle. The difference between different d_y cases are also smaller at $\theta = 75^\circ$.

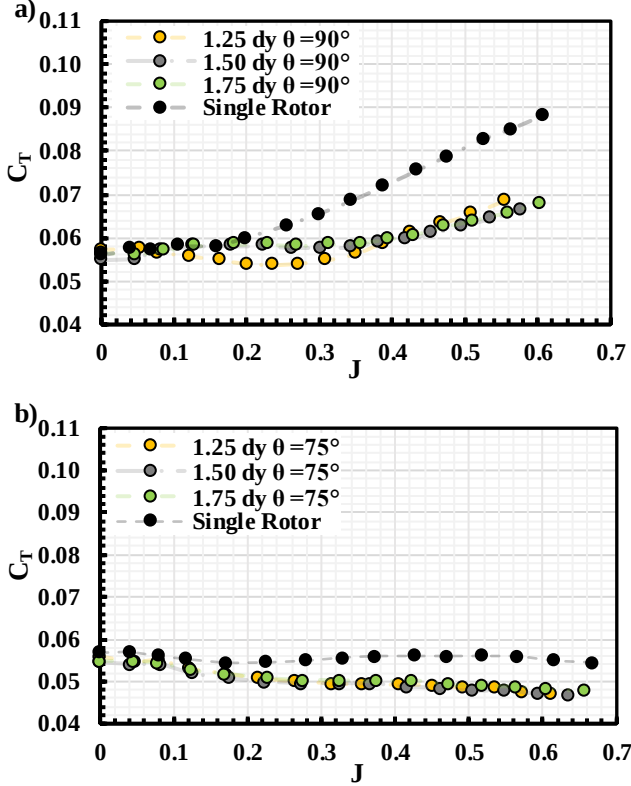


Figure 7. Back Propeller C_T and vs. J under various d_y at a) $\theta = 90^\circ$ and b) $\theta = 75^\circ$

Figure 8 shows the C_{Qx} v.s J for all cases in Figure 8. Different from the changes in C_T , a higher C_{Qx} when compared to the single propeller is observed for all cases at $\theta = 90^\circ$ between $0.1 < J < 0.5$. The difference between different d_y cases is relatively small, except for the $d_y = 1.25$ case at $J < 0.3$ where a smaller C_{Qx} is measured. At $\theta = 75^\circ$, the trend is reversed where a lower C_{Qx} is measured when compared to the single propeller. At $0.1 < J < 0.5$, a smaller d_y results in a smaller measured C_{Qx} indicating that the back propeller is more likely to pitch down due to the interference with the front propeller.

B. Tandem Propeller Performance in Time-Varying Sinusoidal Freestream

The steady freestream data discussed in the previous section is used to create a steady-state model to compare with

the time-varying freestream experiments. The coefficient on each axis was input to a two-dimensional linear interpolation, which accounts for the changes in propeller Reynolds number and advance ratio. The propeller rotational speed and the instantaneous freestream velocity are measured under time-varying freestream conditions, from which the instantaneous advance ratio is calculated. To account for the phase lag between the measured and predicted coefficients due to hotwire installation, the distance between the hotwire and the propeller hub, and the mean velocity of the sinusoidal freestream are used to correct the phase lag. The corrected propeller instantaneous advance ratio is then input into the previously developed 2-D linear interpolation between each data point to get the predicted steady-state propeller response, which can be written as:

$$C_{T_{Predicted}} = C_{T_{Steady}}(J_{measured}) \quad (2.1)$$

$$C_{Q_{Predicted}} = C_{Q_{Steady}}(J_{measured}) \quad (2.2)$$

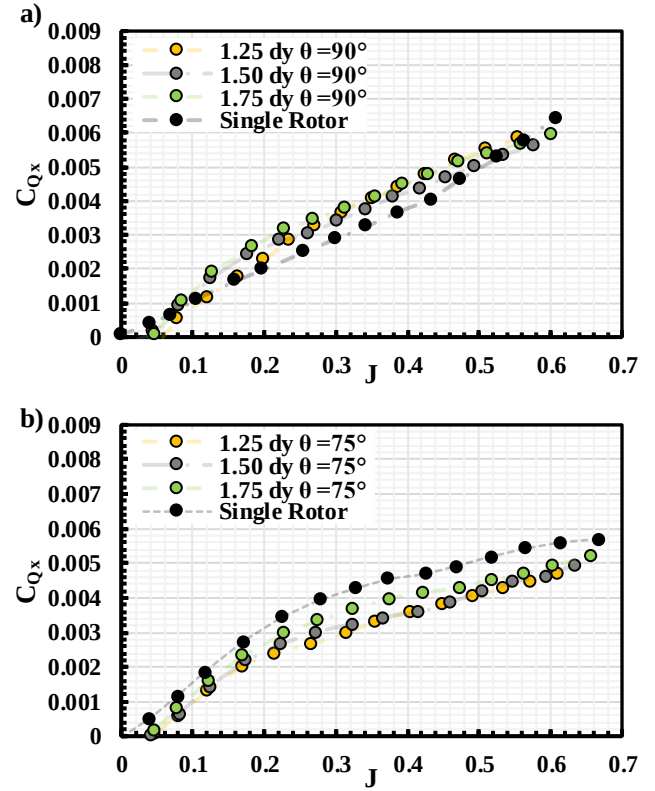


Figure 8. Back Propeller C_{Qx} and vs. J under various d_y at a) $\theta = 90^\circ$ and b) $\theta = 75^\circ$

B.I Tandem propeller performance at $\theta = 90^\circ$

Figure 9 shows the measured and predicted thrust coefficients, $C_{T,m}$ (marked in red) and $C_{T,p}$ (marked in blue), for the front propeller at 67 RPS, $\theta = 90^\circ$ and $\Omega = 300$ RPM. The measured freestream velocity is shown on the right Y-axis. The resulting reduced frequencies $k_\omega = 0.012$, which represent a quasi-steady condition. When the freestream velocity is reduced, the instantaneous propeller advance ratio falls, resulting in an increment in C_{Tz} , as seen

in Figure 6 for the steady freestream response. The steady-state model overlaps the measured C_T at $k_\omega = 0.012$, implying that the response of the propeller is quasi-steady. That is, the propeller responds to the change in advance ratio directly. This trend holds for all front propeller cases tested.

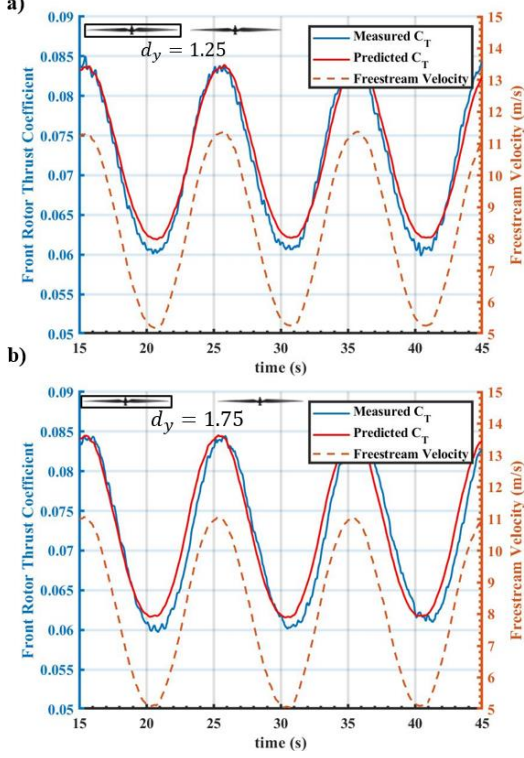


Figure 9. Front propeller measured and predicted C_T at $k_\omega = 0.012$ and $\theta=90^\circ$ for a) $d_y = 1.75$ and b) $d_y = 1.25$

Figure 10 shows the changes in C_T for both d_y cases at a higher k_ω of 0.294, which represents an unsteady condition. A phase lag is observed between the measured value and the steady-state model. A lower peak in the C_T response is also measured when compared to the steady-state model. Both phenomena indicate an unsteady response of the propeller. When reducing the propeller downstream displacement (d_y), a slightly lower C_T along with a slightly higher phase lag are observed for the front propeller. However, the overall changes in the front propeller response are relatively small.

Figure 11 shows the C_{Qx} response of the propeller under the same conditions as Figure 10. Again, a phase lag and a small change in the measured magnitude are observed due to the unsteady response of the propeller at $k_\omega = 0.294$. Overall, the response of the front propeller is similar to the single propeller performance under sinusoidal freestream in our previous study [Ref. 13].

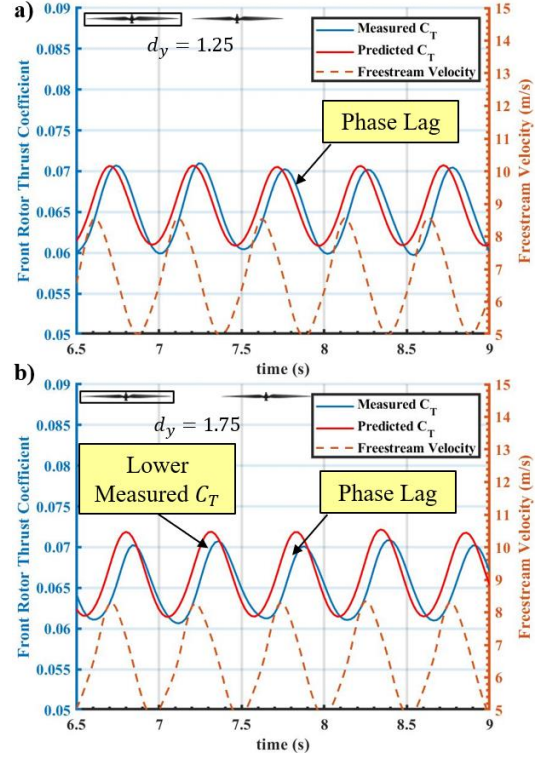


Figure 10. Front propeller measured and predicted C_T at $k_\omega = 0.296$, $\theta=90^\circ$ for a) $d_y = 1.75$ and b) $d_y = 1.25$

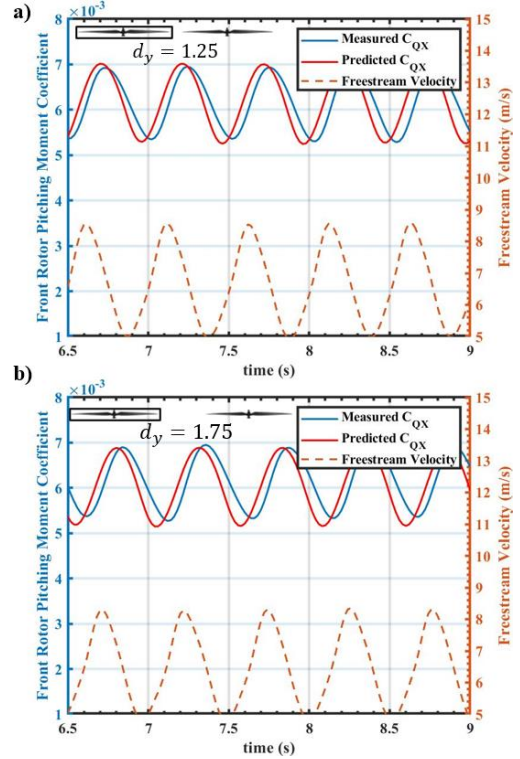


Figure 11. Measured and predicted front propeller C_{Qx} at $k_\omega = 0.296$, $\theta=90^\circ$ for a) $d_y = 1.75$ and b) $d_y = 1.25$

The changes in C_T for the back propeller under the same conditions as Figure 10 is shown in Figure 12. Recall from the steady freestream results in Figure 7, the changes in the back

propeller C_T v.s J is very small. This is reflected in the small variation of the predicted C_T curve based on the steady state performance. At $d_y = 1.75$, a higher variation in C_T is observed when compared to the steady state predictions. Reducing the propeller displacement resulting a smaller variation in C_T for the back propeller. This is likely due to the reduction in the propeller unsteady response causing by the propeller interaction. When placing the propeller in close proximity to each other, the induced velocity interaction reduces the effect of the sinusoidal gust. Moreover, a higher measured C_T is also observed for the $d_y = 1.25$ case when compared to the steady state prediction due to the rotor interaction.

Figure 13 shows the changes in C_{Qx} for the cases in Figure 12. Similar to the response of the front rotor, a clear phase lag and changes in the measured magnitude are observed. For different d_y cases, the measured mean C_{Qx} increases when reducing the propeller displacement, as a result of the changes in the rotor interaction at different propeller displacements.

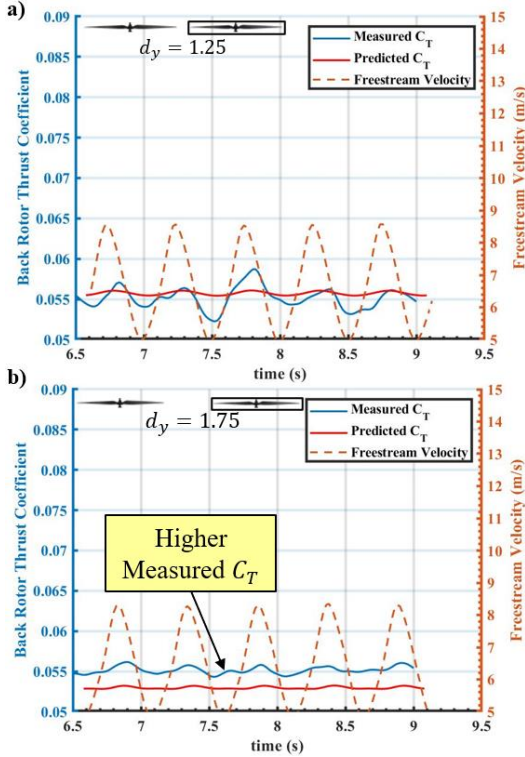


Figure 12. Measured and predicted back propeller C_T at $k_\omega = 0.296$, $\theta=90^\circ$ for a) $d_y = 1.75$ and b) $d_y = 1.25$

To better represent the phase lag of the propeller response, the instantaneous measured C_T for both front and back propeller v.s J is shown in Figure 15. The steady-state performance is also shown in the same plot as the dashed line. A clear hysteresis loop occurred for the measured C_T of the front propeller as a result of the phase lag in the propeller thrust response. However, the hysteresis loop is not quite clear for the results of the back propeller, as a result of a smaller variation of C_T v.s J for the back propeller as seen in Figure 7 a. It is also worth noting that when reducing the propeller

displacement, the magnitude of the measured C_T increases while the variation reduces, which is also reflected in Figure 12.

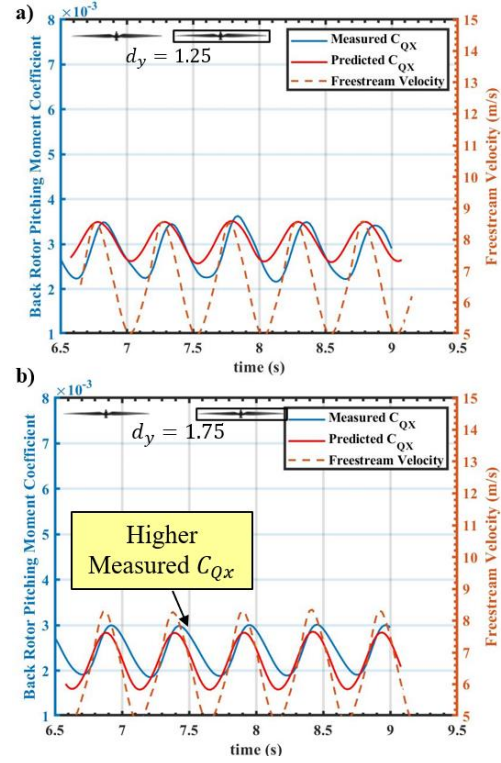


Figure 13. Measured and predicted back propeller C_{Qx} at $k_\omega = 0.296$, $\theta=90^\circ$ for a) $d_y = 1.75$ and b) $d_y = 1.25$

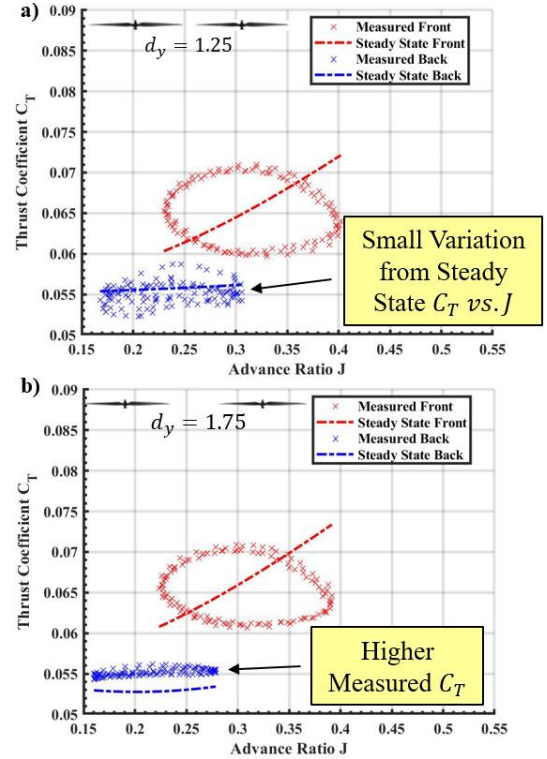


Figure 14. Measured and steady state C_T vs. J at $k_\omega = 0.296$, $\theta = 90^\circ$ for a) $d_y = 1.75$ and b) $d_y = 1.25$

As the changes in C_{Qx} vs. J for both propellers is more significant than C_T , the results can provide a better representation of the phase lag for both propellers. The measured C_{Qx} for both front and back propeller v.s J is shown in Figure 15 at $k_\omega = 0.151$. Similarly, a hysteresis loop occurred for the measured C_{Qx} for both propellers indicating a phase lag for the C_{Qx} response. A slightly larger hysteresis loop for the front propeller is observed for the $d_y = 1.75$ when compared to $d_y = 1.25$ case, indicating a slightly higher phase lag of the propeller response. This trend holds for the back propeller performance. Although placing the propeller at small distance results in the back propeller performance loss at the steady state, the propeller interaction seems to reduce the unsteady response of both the front and the back propeller. It is also worth noting that the measured C_{Qx} for $d_y = 1.25$ is higher than the steady state model as a result of a higher measured C_T when compared to the steady state model for the back propeller seen in Figure 12 and Figure 14.

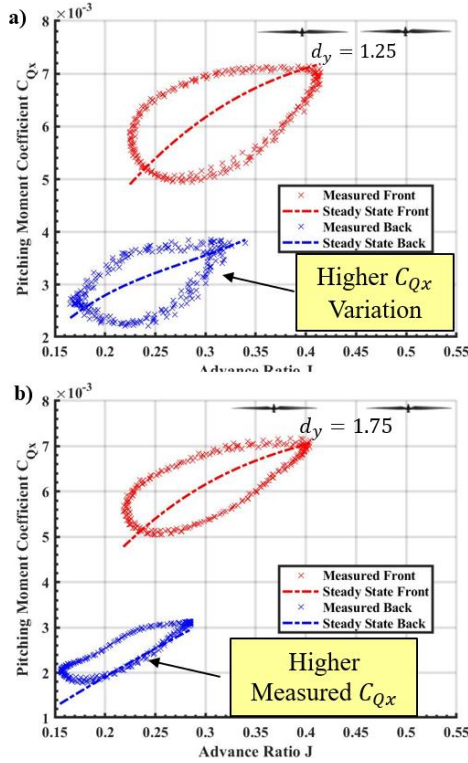


Figure 15. Measured and steady state C_{Qx} vs. J at $k_\omega = 0.151$, $\theta = 90^\circ$ for a) $d_y = 1.75$ and b) $d_y = 1.25$

Figure 16 shows the instantaneous measured C_{Qx} for both front and back propeller at a higher reduced frequency of 0.296. A larger hysteresis loop is observed for both propellers at both d_y cases, indicating a higher phase lag in the propeller response when increasing the reduced frequency, as expected. The overall trend between the $d_y = 1.25$ and $d_y = 1.75$ cases is similar to ones seen in Figure 15.

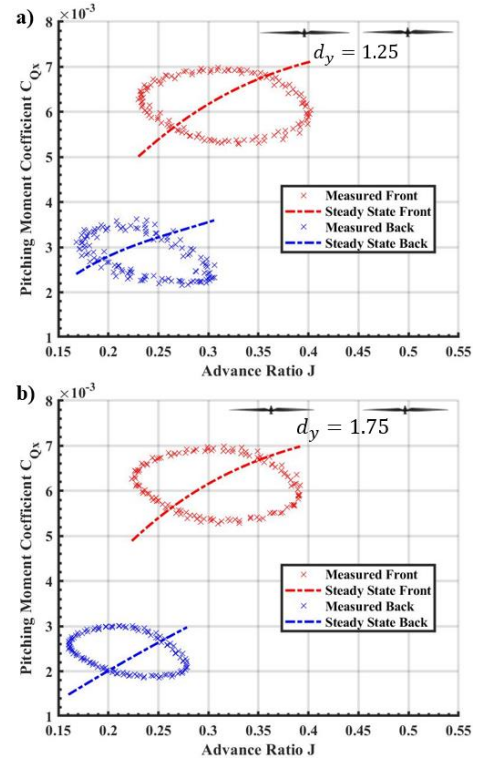


Figure 16. Measured and steady state C_{Qx} vs. J at $k_\omega = 0.296$, $\theta = 90^\circ$ for a) $d_y = 1.75$ and b) $d_y = 1.25$

Figure 17 summarizes the overall difference between the measured and steady state C_T for all d_y cases at various k_ω . The mean value is marked as discrete data points while the upper and lower bound of the coefficient variation is marked as dashed lines. To calculate the variation bound, each peak and trough in the measured data under each sinusoidal wave is calculated. The value is then averaged among all of the peak and trough values in each test trial to get the upper and lower bound.

For the front propeller, at $k_\omega \leq 0.2$, the mean value of the measured C_T matches with the steady state model relatively well as the datapoint overlaps, except for the $d_y = 1.25$ case where a lower C_T is measured due to propeller interference. At $k_\omega > 0.2$, a small reduction in C_T is observed when compared to the steady-state model. A similar result is also observed in our previous study [Ref. 13] for the single rotor performance, as a result of the propeller's unsteady response. For the back propeller, the difference between the measured mean and the steady state model is small at $d_y \geq 1.5$ due to the small variation in C_T . At $d_y = 1.25$, the measured mean is higher than the steady-state model across all k_ω , which is also reflected in Figure 12. It is worth noting that with the decrement in propeller displacement distance, the variation in back propeller C_T reduces, which also agrees with the trends observed in Figure 12.

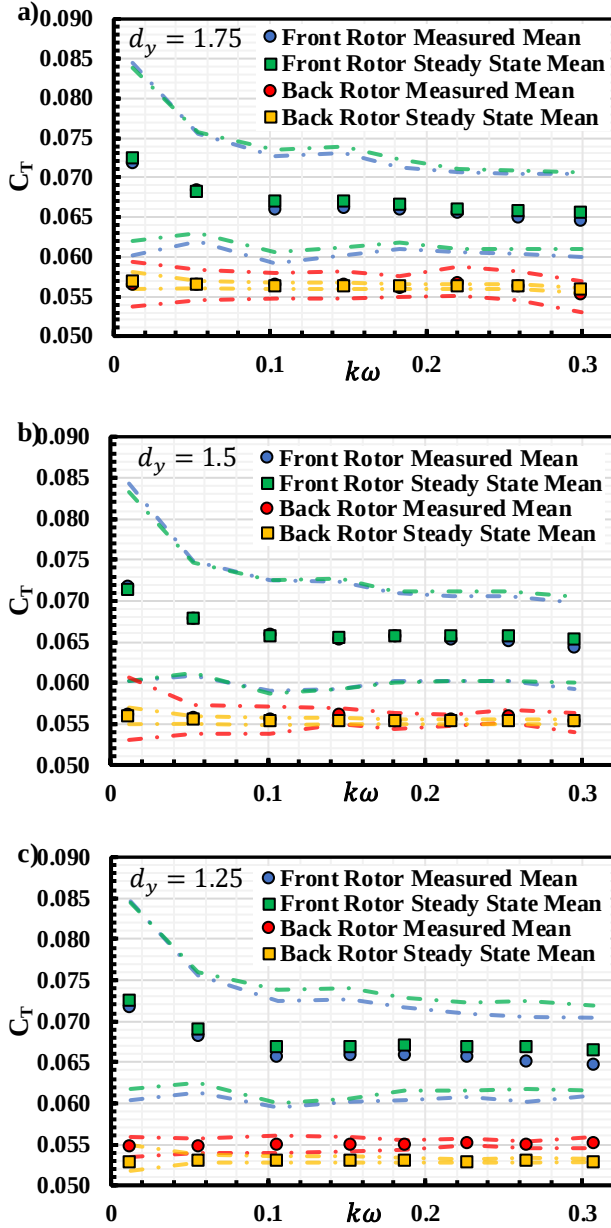


Figure 17. Measured and Steady State C_T vs. k_ω at $\theta = 90^\circ$ for a) $d_y = 1.75$, b) $d_y = 1.50$ and c) $d_y = 1.25$

Figure 18 shows the mean measured and steady state C_{Qx} for both propellers along with the upper and lower bound for C_{Qx} at all d_y cases tested. The difference in the front rotor C_{Qx} between the measured and steady-state model is almost negligible. While for the back propeller at $d_y \geq 1.5$, a small reduction in measured C_{Qx} is observed at $k_\omega > 0.2$, which is similar to the trend observed in our previous study [Ref. 13] for the single propeller performance. When reducing the propeller displacement to $d_y = 1.25$, an increment in C_{Qx} for the back propeller is observed at $k_\omega > 0.15$, which is a reversed trend when compared to the higher d_y cases. Overall, the back propeller will generate higher thrust and trend to pitch up more when placed in close proximity to the front propeller under sinusoidal freestream.

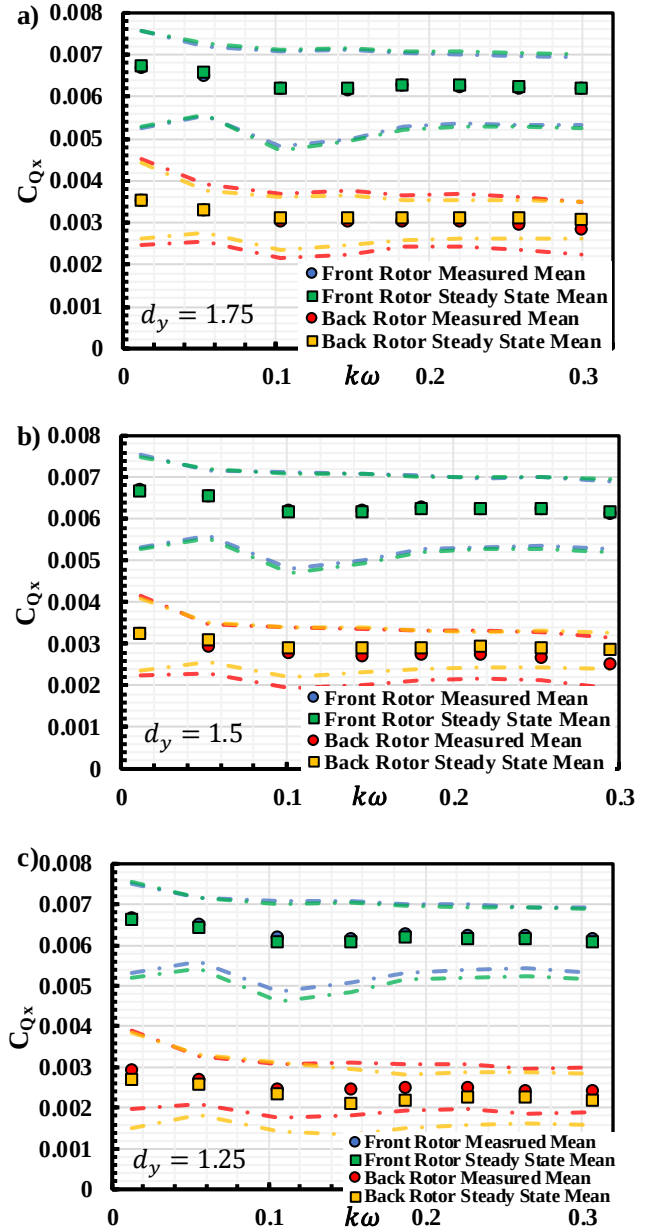


Figure 18. Measured and Steady State C_{Qx} vs. k_ω at $\theta = 90^\circ$ for a) $d_y = 1.75$, b) $d_y = 1.50$ and c) $d_y = 1.25$

To account for the overall stability of the tandem propeller system, the overall pitching moment of the system is calculated based on the measured value and the steady state model using Equation 2 below.

$$C_{Qxsys} = C_{QxF} + C_{QxB} + (T_{Front} - T_{Back}) * \frac{0.5d}{\rho n^2 D^5} \quad (2)$$

The results for all d_y cases tested are shown in Figure 19. For all cases at $k_\omega > 0.15$, the measured C_{Qxsys} is lower than the steady state model, as a result of the changes in the C_T of both propellers and changes in the C_{Qx} of the back propeller. This decrement in C_{Qxsys} increases when decreasing d_y , especially for the $d_y = 1.25$ case. Indicating a reduction in

the tandem propeller system stability in time-varying freestream when placing the propeller closer to each other.

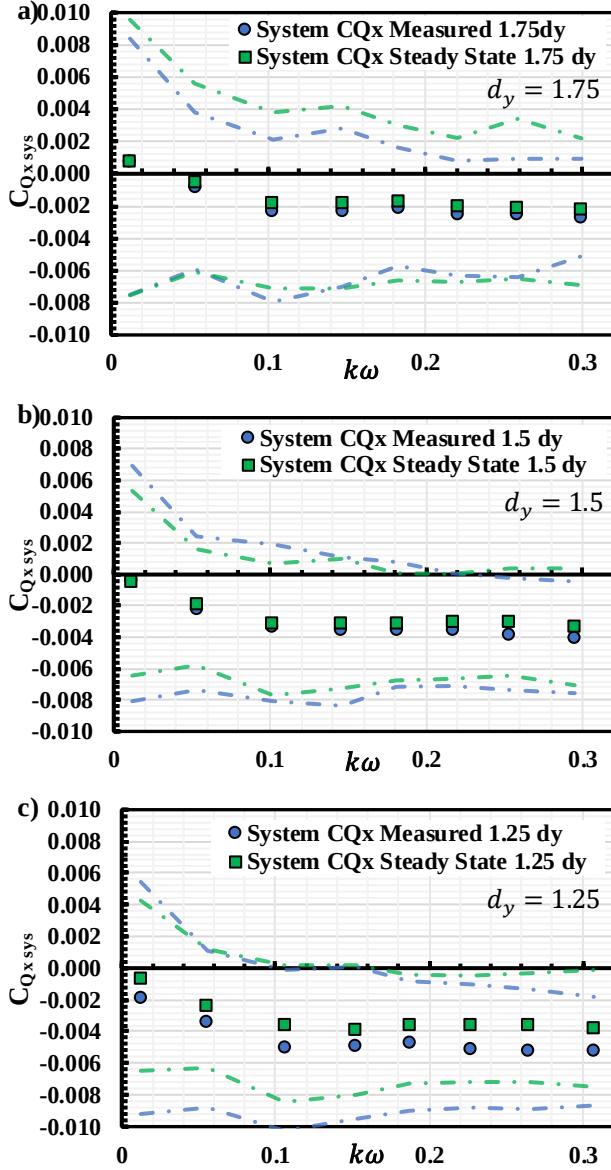


Figure 19. Measured and Steady State $C_{Qx\ sys}$ mean and variation vs. k_ω at $\theta = 90^\circ$ for a) $d_y = 1.75$, b) $d_y = 1.50$ and c) $d_y = 1.25$

The overall phase lag, $\Delta\phi$, of C_T , C_{Qx} and $C_{Qx\ sys}$ at $\theta = 90^\circ$ is shown in Figure 20. Due to the unique propeller response under streamwise gust, the traditional phase lag calculation which uses the positive zero passing point is not suitable.

Hence, an alternative method is used. To do this, a sinusoidal wave is fitted to the measured data and the steady state model after the correction of installation phase lag. To find the best fit sinusoidal wave, the root-mean-square (RMS) value between the fitted sinusoidal wave and the experimental data is calculated. The wave with the lowest RMS is considered the best-fit wave. The phase difference between the best fit sinusoidal wave of the measured data

and the steady state model represents the phase lag.

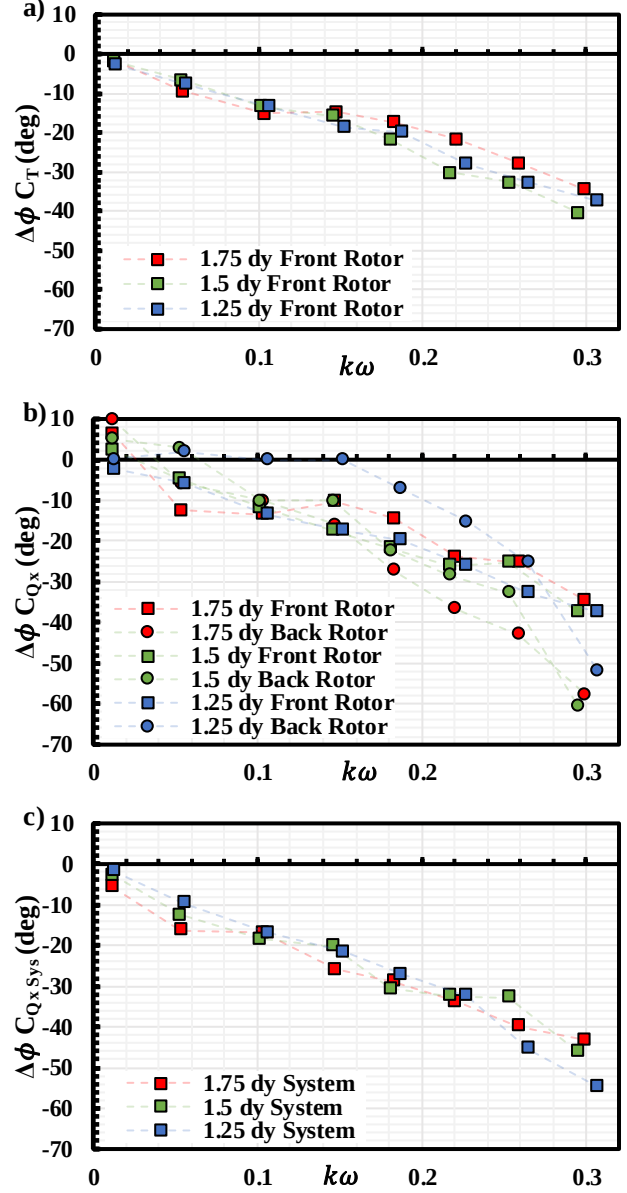


Figure 20. Phase lag ($\Delta\phi$) of a) C_T , b) C_{Qx} and c) $C_{Qx\ sys}$ at $\theta = 90^\circ$

Due to the small variation in C_T for the back propeller seen in Figure 14 and Figure 17, the phase lag is difficult to be identified for the back propeller. Therefore, only the result from the front propeller is shown. With the increment in k_ω , the phase lag in C_T increases as expected. The difference in $\Delta\phi_{C_T}$ at different propeller placement is relatively small. For $\Delta\phi_{C_{Qx}}$ in Figure 20.b, the overall trend seen in the $\Delta\phi_{C_T}$ is preserved. Again, the difference on the front rotor $\Delta\phi_{C_{Qx}}$ is relatively small. However, with the increment in d_y , a clear difference in $\Delta\phi_{C_{Qx}}$ for the back propeller is observed. A smaller $\Delta\phi_{C_{Qx}}$ is found when placing the propeller closer to each other. For the phase lag of the system response, $\Delta\phi_{C_{Qx\ sys}}$, the overall trend is also similar to $\Delta\lambda_{C_T}$ where the

difference between different propeller displacement is small. In this case, despite the difference seen in $\Delta\lambda_{C_{Qx}}$, the overall system response is similar among all d_y cases.

B.II Tandem propeller performance at $\theta = 75^\circ$

The performance of the tandem propeller at $\theta = 75^\circ$, which represents a forward flight condition, is discussed in this section. Figure 21 shows the instantaneous measured C_T for both front and back propellers *vs.* J at the highest k_ω tested. The hysteresis loop is observed for both propellers. However, due to the small variation in C_T *vs.* J for the front propeller, the hysteresis loop is less clear when compared to the $\theta = 90^\circ$ cases. It is worth noting that the C_T response for the back propeller is approximately 90° off phase as the hysteresis loop has an opposite slope when compared to the steady state response. When reducing the propeller displacement, $\Delta\phi_{C_T}$ for the back propeller reduces, as the overall slope difference between the hysteresis loop and the steady state response is smaller for the $d_y = 1.25$ case when compared to the $d_y = 1.75$ case.

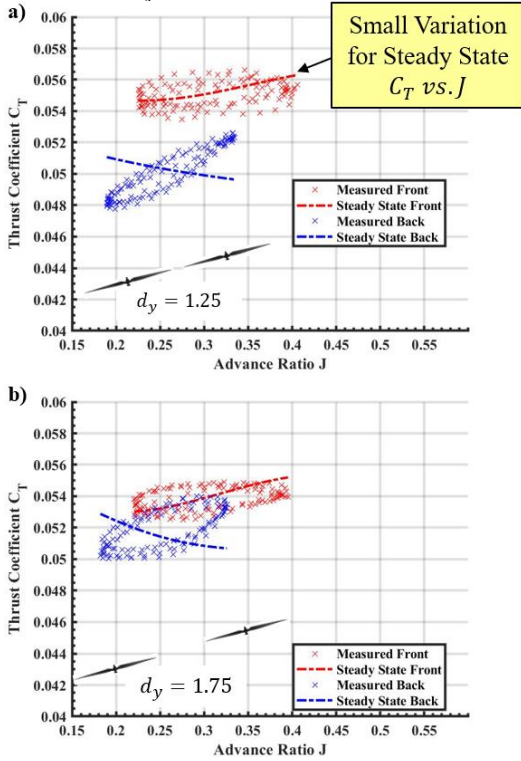


Figure 21. Measured and steady state C_T *vs.* J at $k_\omega = 0.294$, $\theta = 75^\circ$ for a) $d_y = 1.75$ and b) $d_y = 1.25$

Figure 22 shows the instantaneous measured C_T for both front and back propellers *vs.* J for the cases shown in Figure 21. Due to the higher variation in C_{Qx} *vs.* J , the hysteresis loop is very clear. The difference in $\Delta\phi_{C_{Qx}}$ for the front propeller at different d_y is relatively small while a lower $\Delta\phi_{C_{Qx}}$ for the back propeller is observed at smaller propeller displacement, which agrees with the observation in Figure 21.

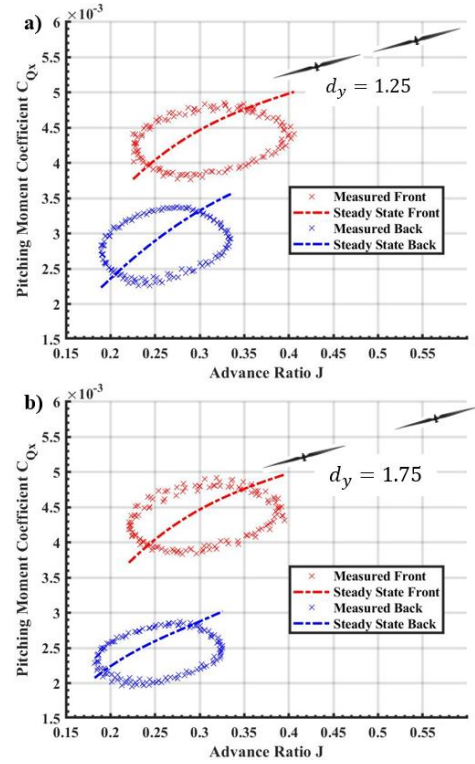


Figure 22. Measured and steady state C_{Qx} *vs.* J at $k_\omega = 0.294$, $\theta = 75^\circ$ for a) $d_y = 1.75$ and b) $d_y = 1.25$

Figure 23 shows the C_T for both propellers at all d_y cases tested. At $d_y \geq 1.5$, the mean measured C_T for both front and back propeller agrees with the steady state model well among all k_ω range tested, which agrees with the results for the $\theta = 90^\circ$ case. For the smallest propeller displacement where $d_y = 1.25$, an increase in the back propeller measured mean C_T and a decrease in the front propeller measured mean C_T is observed when compared to the steady-state model due to the propeller interference. It is worth noting that the latter phenomena are not observed in the $\theta = 90^\circ$ case.

Figure 24 shows the measured and steady state C_{Qx} for all cases in Figure 23. Surprisingly, the measured value overlaps with the steady state model at $\theta = 75^\circ$, indicating a reduction of the pitching moment magnitude change in forward flight when encountering the gust.

The overall pitching moment for the tandem propeller system is shown in Figure 25 for all cases tested. Similar to $\theta = 90^\circ$ cases, the measured data and the steady state model agree relatively well. Although a small difference between the measured and the steady state $C_{Qx Sys}$ is observed at $k_\omega > 0.15$, the overall difference in the magnitude is smaller than the $\Delta\phi_{C_{Qx Sys}}$ observed at the $\theta = 90^\circ$ cases. In general, when placing the front and back propeller close to each other, the tandem propeller system will tend to pitch down more when encountering the streamwise gust in edgewise flight. Especially at higher forward flight θ for the tandem propeller system.

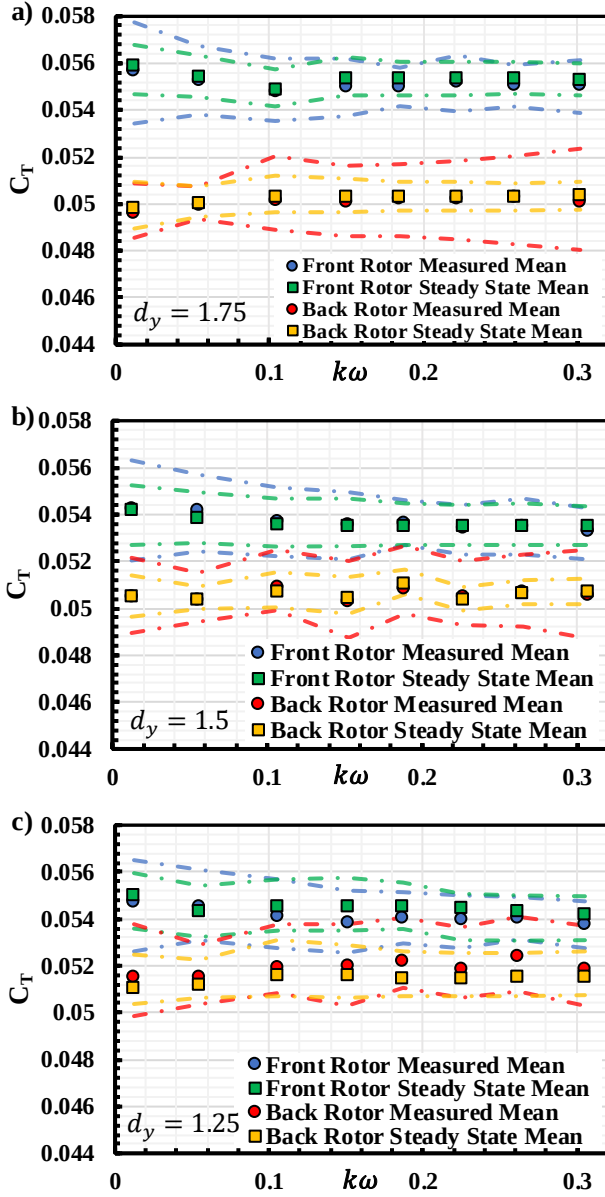


Figure 23. Measured and Steady State C_T vs. $k\omega$ at $\theta = 75^\circ$ for a) $d_y = 1.75$, b) $d_y = 1.50$ and c) $d_y = 1.25$

Figure 26 shows the overall phase lag of all coefficients using the same method as the ones in Figure 20. Unlike the $\theta = 90^\circ$ case, significant difference in $\Delta\phi$ is found between the front and back propellers at $\theta = 75^\circ$. $\Delta\phi$ for the front propeller C_T is relatively small, which agrees with the trend for the single propeller seen in our previous study [Ref. 13]. However, $\Delta\phi$ for the back propeller increases with the increment of $k\omega$, with a much higher magnitude when compared to the $\Delta\phi$ for the front propeller. Moreover, a lower d_y would result in a smaller C_T phase lag in both the front and the back propeller, which is also seen in the C_{Qx} results at $\theta = 90^\circ$ in Figure 20.

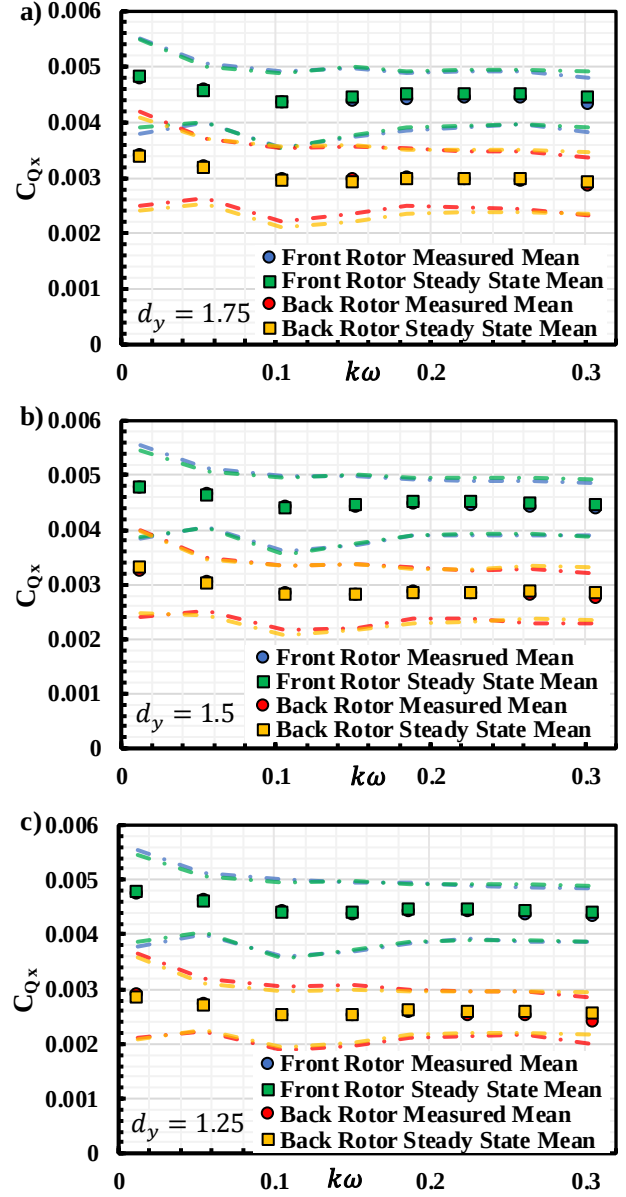


Figure 24. Measured and Steady State C_{Qx} vs. $k\omega$ at $\theta = 75^\circ$ for a) $d_y = 1.75$, b) $d_y = 1.50$ and c) $d_y = 1.25$

For the phase lag in C_{Qx} , the difference in the front propeller at different d_y is relatively small, while the overall change in magnitude is similar to the front propeller performance at $\theta = 90^\circ$. This also agrees with the results for the single propeller from our previous study [Ref. 13]. For the back propeller, a phase lead in the C_{Qx} response is observed at $J < 0.2$. The phase lag in the back propeller then increases at $J > 0.2$ and trends towards the response of the front propeller. Similar to the result for C_T , a smaller propeller displacement (d_y) would result in a lower phase lag, or a higher phase lead in the back propeller C_{Qx} response.

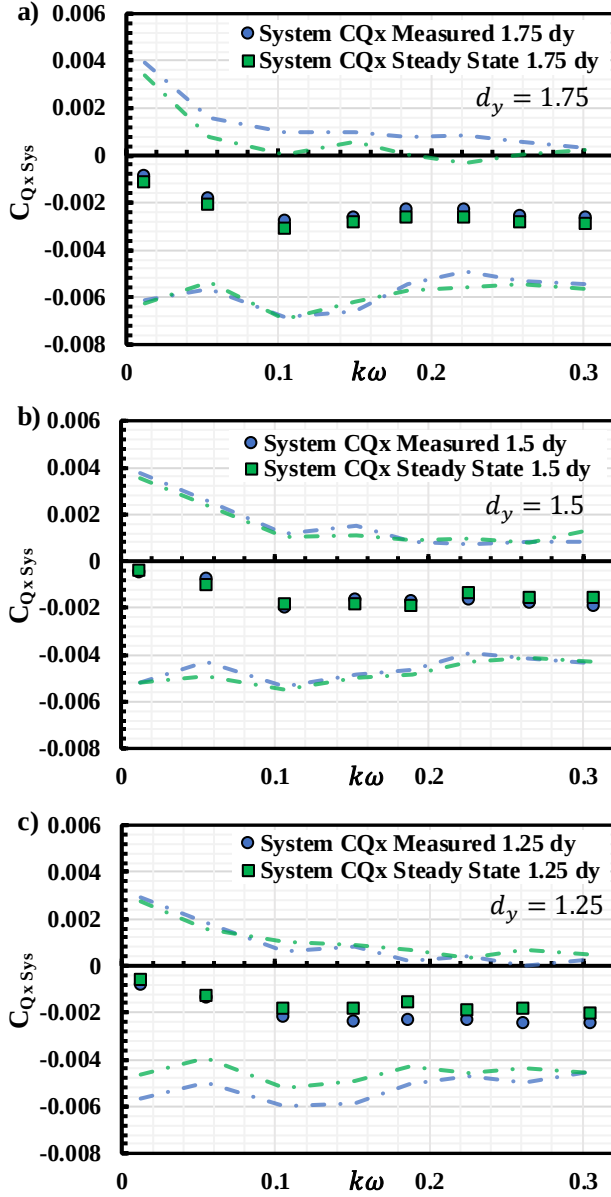


Figure 25. Measured and Steady State C_{QxSys} vs. $k\omega$ at $\theta = 75^\circ$ for a) $d_y = 1.75$, b) $d_y = 1.50$ and c) $d_y = 1.25$

For the $\Delta\phi$ of the overall tandem propeller system, due to the higher phase lag observed in both front and back propeller at higher d_y , the overall phase lag of the tandem propeller system is also higher at larger propeller displacement. Again, although placing the front and the back propeller in close proximity will result in a smaller phase lag of the system, a reduction in the mean C_{QxSys} and an increase in the C_{QxSys} variation is also observed. This is to say, the system is not more stable when placing the propeller in close proximity. On contrary, the tandem propeller system's response to the gust is faster with a higher likelihood of pitching downward when placing the propeller in close proximity, which increases the requirement of the drone control system to maintain a stable flight under gust encounter. Thus, placing the back propeller

at least $0.5D$ away from the front propeller will reduce the instability when encountering the streamwise gust.

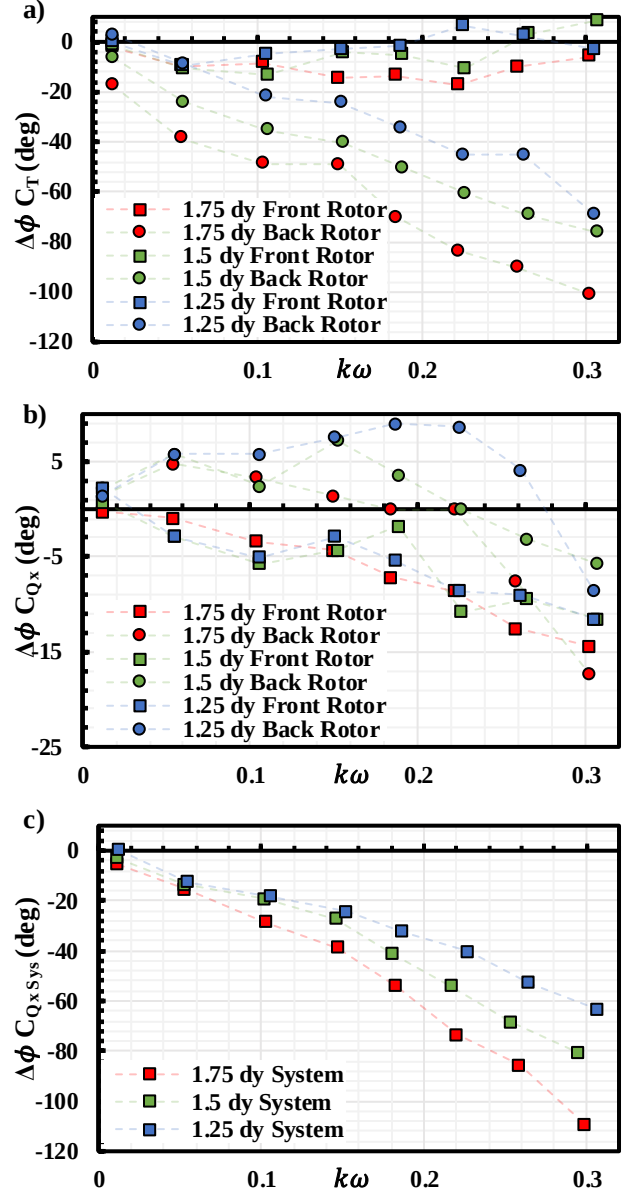


Figure 26. Phase lag ($\Delta\phi$) of a) C_T , b) C_{Qx} and c) C_{QxSys} at $\theta = 75^\circ$

CONCLUSIONS

The aerodynamic performance of the tandem propeller system under steady and time-varying freestream is investigated with the wind tunnel experiment. Under the steady freestream conditions, placing the front and back propeller in close proximity to each other resulted in an increase in the front propeller thrust generation and a decrease in the back propeller thrust generation, which agrees with the trend observed in the literature. The effect on the front propeller pitching moment is almost negligible, while the back propeller pitching moment increases at a higher

incidence angle and decreases at a lower incidence angle in close proximity to the front propeller.

The experimental result for the tandem propeller system under the steady freestream was used to build a steady-state model and compared to the experimental results under the time-varying freestream. For both incidence angles tested, the front propeller performance resembles the single propeller performance in both change in coefficient magnitude and phase lag at large propeller displacement. A reduction in the front propeller thrust is observed when placing the propeller in close proximity to each other. The steady-state model did a relatively good job of predicting the performance of the front propeller, as well as predicting the pitching moment of each individual propeller at all reduced frequencies tested in the current study.

However, the overall performance of the back propeller differs from the performance of the front propeller in both magnitude and phase lag. An increment in thrust coefficient is observed for the back propeller when placing it in close proximity to the front propeller under streamwise gust. On the other hand, a decrement in the front propeller thrust coefficient is also observed. Moreover, the overall tandem rotor system trend to pitch down further when encountering the streamwise gust.

The phase lag for the back propeller thrust coefficient is much higher than that of the front propeller, especially at lower forward flight θ . While the phase lag in the pitching moment is relatively smaller under all θ tested. Placing the propeller in close proximity to each other will reduce the phase lag of both front and propeller response, resulting in a faster response of the tandem rotor system when encountering the streamwise gust. Combined with the higher pitch-down moment the tandem propeller system experienced at small propeller displacement, such a design would introduce additional control complications for the drone applications when encountering the streamwise gust.

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